# SIMATS ENGINEERING
## ASSESSMENT TOOL 1: Analytical Problem Solving

**COURSE NAME: COMPUTER VISION with OpenCV**
**COURSE CODE: DSA0204**

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**COURSE OUTCOME COVERED**

**CO2:** *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

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### 1. Grey Level Transformation

**(a)** Explain the purpose of grey level transformation in image enhancement.
**(b)** Apply a negative transformation $s = 255 - r$ to the pixel values:

$$r = [50,100,150,200]$$

Compute the transformed values.

### 2. Log Transformation

**(a)** Explain how log transformation enhances low-intensity pixels.
**(b)** Given $s = clog(1 + r)$, where $c = 10$, compute $s$ for:

$$r = [0,10,50,100]$$

### 3. Histogram Processing

**(a)** Explain the role of histogram equalization in contrast enhancement.

**(b)** Given a 3-bit image with histogram:

| Intensity | Frequency |
|---|---|
| 0 | 2 |
| 1 | 2 |
| 2 | 4 |

| Intensity | Frequency |
| --- | --- |
| 3 | 8 |

Compute the cumulative distribution and equalized intensity values.

**4. Spatial Filtering (Smoothing)**

**(a)** Explain the purpose of spatial filtering for smoothing.

**(b)** Apply a 3×3 averaging filter to the following region:

$$[10\ 20\ 30\ 20\ 30\ 40\ 30\ 40\ 50\ ]$$

Compute the new center pixel value.

**5. Spatial Filtering (Sharpening)**

**(a)** Explain how sharpening enhances image details.

**(b)** Apply the Laplacian mask:

$$[[0\ -1\quad 0\},\{-1\quad 4\ -1],\ \{\ 0\ -1\quad 0\ ]]$$

to the matrix:

$$[10\quad 10\quad 10, 10\quad 20\quad 10, 10\quad 10\quad 10\ ]$$

Compute the output at the center.

**6. Frequency Domain Filtering**

**(a)** Differentiate low-pass and high-pass filtering in frequency domain.

**(b)** Given frequency components:

$$F = [100,80,40,20]$$

Apply a low-pass filter that retains only the first two components. Write the filtered output.

**7. Thresholding**

**(a)** Explain global thresholding in segmentation.

**(b)** Given pixel values:

$$[200]$$

Apply threshold $T = 100$. Assign values (0 or 1).

**8. Edge Detection**

**(a)** Explain the concept of edge detection.

**(b)** Apply the Sobel operator (horizontal):

$$[-1\ -2\ -1,\quad 0\quad 0\quad 0,\ 1\quad 2\quad 1\ ]$$

to:

$$[10\ \ 10\ \ 10,\ \ 20\ \ 20\ \ 20,\qquad 30\ \ 30\ \ 30\,]$$

Compute the gradient at the center.

### 9. Region-Based Segmentation

**(a)** Explain region growing in segmentation.

**(b)** Given pixel intensities:

$$[1\ \ 0\ \ 2]$$

If seed = 50 and threshold difference = 5, identify pixels belonging to the region.

### 10. Combined Enhancement & Segmentation

**(a)** Explain why smoothing is applied before segmentation.

**(b)** Given image values:

$$[1\ \ 0\ \ 5]$$

First apply averaging (assume mean = 48), then threshold at $T = 50$.

Determine final segmented output.

### 11. Power-Law (Gamma) Transformation

A grayscale image is enhanced using power-law transformation.

(a) Interpret the role of gamma correction in image enhancement.

(b) Justify the use of transformation $s = cr^{\gamma}$, where $\gamma = 0.5$, $c = 1$.

(c) Apply the transformation step-by-step to pixel values:

$r = [25,\ 100,\ 225]$.

(d) Compute the transformed values accurately.

(e) Interpret the results in terms of brightness enhancement.

### 12. Contrast Stretching

An image has intensity values in the range [50, 150].

(a) Interpret the need for contrast stretching.

(b) Justify linear transformation to range [0, 255].

(c) Apply transformation step-by-step for values:

$r = [50,100,150]$.

(d) Compute the output values accurately.

(e) Interpret improvement in contrast.

### 13. Histogram Equalization (4-bit Image)

A 4-bit image has the following histogram:

| Intensity | Frequency |
|---|---|
| 0 | 4 |
| 1 | 6 |
| 2 | 10 |
| 3 | 6 |

(a) Interpret the need for equalization.

(b) Justify use of CDF.

(c) Compute cumulative distribution.

(d) Compute new intensity levels.

(e) Interpret contrast enhancement.

**14. Median Filtering**

A noisy image contains salt-and-pepper noise.
(a) Interpret noise characteristics.
(b) Justify median filter usage.
(c) Apply filter to:

$$10 \quad 200 \quad 10, 200 \quad 50 \quad 200, 10 \quad 200 \quad 10$$

(d) Compute center value.

(e) Interpret noise removal effectiveness.

**15. Gaussian Smoothing**

An image is smoothed using Gaussian filter.
(a) Interpret smoothing objective.
(b) Justify Gaussian filter over averaging.
(c) Apply filter:

$$[1 \quad 2 \quad 1, \; 2 \quad 4 \quad 2, \; 1 \quad 2 \quad 1]$$

to matrix:
$[10 \quad 20 \quad 10, 20 \quad 40 \quad 20, 10 \quad 20 \quad 10]$
(d) Compute center pixel.

(e) Interpret smoothing effect.

**16. High-Pass Filtering**

An image requires sharpening.
(a) Interpret need for high-pass filtering.
(b) Justify filter selection.

(c) Apply mask:
$[-1 \quad -1 \quad -1, -1 \quad 8 \quad -1, -1 \quad -1 \quad -1]$
(d) Compute center value.

(e) Interpret edge enhancement.

**17. Frequency Domain High-Pass**

Given frequency components:

$$F = [10,20,80,100]$$

(a) Interpret high-frequency importance.

(b) Justify high-pass filtering.

(c) Apply filtering (remove low frequencies).

(d) Compute output.

(e) Interpret result.

**18. Adaptive Thresholding**

Pixel values: [60, 80, 120, 140]

(a) Interpret segmentation challenge.

(b) Justify adaptive thresholding.

(c) Apply local threshold (mean = 100).

(d) Compute binary output.

(e) Interpret result.

**19. Edge Detection (Vertical Sobel)**

Apply vertical Sobel:

$$[-1 \quad 0 \quad 1, -2 \quad 0 \quad 2, -1 \quad 0 \quad 1]$$

Matrix:

$$[10 \quad 20 \quad 30, 10 \quad 20 \quad 30, 10 \quad 20 \quad 30]$$

(a) Interpret edge detection objective.

(b) Justify operator choice.

(c) Apply step-by-step.

(d) Compute center value.

(e) Interpret edge direction.

**20. Prewitt Edge Detection**

(a) Interpret Prewitt operator purpose.
(b) Justify use over Sobel.

(c) Apply kernel:
$[-1 \quad 0 \quad 1, -1 \quad 0 \quad 1, -1 \quad 0 \quad 1]$
(d) Compute output.

(e) Interpret result.

**21. Region Splitting**

Pixel set: [40, 45, 100, 105]

(a) Interpret segmentation problem.

(b) Justify splitting method.

(c) Apply threshold 80.

(d) Compute regions.

(e) Interpret separation.

**22. Region Merging**

Regions: [45, 50], [52, 55]

(a) Interpret merging criteria.

(b) Justify merging approach.

(c) Apply threshold = 5.

(d) Compute merged region.

(e) Interpret result.

**23. Edge Linking**

Edges detected: [1, 0, 1, 0, 1]

(a) Interpret problem of discontinuity.

(b) Justify linking method.

(c) Apply linking step.

(d) Compute result.

(e) Interpret continuity.

**24. Low-Pass Filtering (Frequency)**

Given $F = [200,150,50,20]$

(a) Interpret smoothing requirement.

(b) Justify LPF selection.

(c) Apply filter.

(d) Compute output.

(e) Interpret effect.

**25. Laplacian Sharpening (Variant)**

Mask:

$$\begin{matrix} 0 & -1 & 0, \\ -1 & 5 & -1, \\ 0 & -1 & 0 \end{matrix}$$

(a) Interpret sharpening need.

(b) Justify mask.

(c) Apply to matrix.

(d) Compute center.

(e) Interpret result.

**26. Histogram Matching**

(a) Interpret need for matching.

(b) Justify method.

(c) Map values [10, 50, 100].

(d) Compute new values.

(e) Interpret.

**27. Noise Reduction Comparison**

(a) Interpret noise type.

(b) Compare averaging vs median.

(c) Apply both.

(d) Compute outputs.

(e) Interpret differences.

**28. Combined Filtering**

(a) Interpret need for smoothing + sharpening.

(b) Justify sequence.

(c) Apply filters stepwise.

(d) Compute output.

(e) Interpret improvement.

**29. Multi-Threshold Segmentation**

Pixels: [20, 60, 120, 180]

(a) Interpret multi-level segmentation.

(b) Justify thresholds (50,100).

(c) Apply segmentation.

(d) Compute classes.

(e) Interpret.

**30. Boundary Detection**

(a) Interpret boundary extraction.

(b) Justify method.

(c) Apply edge + linking.

(d) Compute result.

(e) Interpret boundaries.

***Code of Conduct:***

*I ____Bhavya PS(192524017)_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*


***Signature of the student:***

RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Problem Understanding | 20% | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps | Partial understanding; misses key elements | Misinterprets or unable to understand problem |
| Method Selection | 20% | Selects most appropriate method/formula with strong justification | Selects correct method with minor errors | Inappropriate or partially correct method | Unable to select method |
| Solution Process | 25% | Logical, systematic steps with correct approach throughout | Mostly correct steps with minor mistakes | Disorganized steps; multiple errors | No clear process |
| Accuracy of Calculation | 20% | All calculations accurate; correct final answer | Minor calculation errors | Frequent errors; incorrect final answer | No valid calculations |
| Interpretation of Results | 15% | Clearly interprets results with units and engineering relevance | Basic interpretation with minor omissions | Limited or unclear interpretation | No interpretation |

**Question 1 – Grey Level Transformation**

(a) Explain the purpose of grey level transformation in image enhancement.

Definition

Grey level transformation is a point processing technique in digital image processing where the intensity value of every pixel is modified independently according to a predefined mathematical function.

Purpose

- Improves the visual appearance of an image.
- Enhances contrast between dark and bright regions.
- Highlights important features that may not be clearly visible.
- Corrects poor illumination.
- Prepares images for segmentation and object detection.
- Reduces the effect of uneven lighting.

Applications

- Medical image enhancement (X-rays, MRI scans)
- Satellite image processing
- Fingerprint recognition
- Industrial inspection
- Computer Vision systems

Conclusion

Grey level transformation improves image quality by modifying pixel intensities, making important objects easier to identify.

(b) Apply a negative transformation to the pixel values. Compute the transformed values.

Formula

For an image having L intensity levels,

$$s = (L - 1) - r$$

where

- r = original pixel value
- s = transformed pixel value

For an 8-bit image

$$L = 256$$

Hence,

$$s = 255 - r$$

Example Calculation

Suppose the pixel values are

| Original Pixel (r) | Negative Pixel (s=255−r) |
|---|---|
| 40 | 215 |
| 75 | 180 |
| 120 | 135 |
| 180 | 75 |
| 220 | 35 |

Final Answer

Negative transformed values are

215, 180, 135, 75, 35

Interpretation

Dark pixels become bright, and bright pixels become dark. This is useful for enhancing white details present on dark backgrounds, such as medical X-ray images.

**Question 2 – Log Transformation**

(a) Explain how log transformation enhances low-intensity pixels.

Definition

Log transformation expands dark pixel values and compresses bright pixel values.

Formula

$$s = c\log(1 + r)$$

where

- r = input pixel
- c = scaling constant
- s = output pixel

Working

- Small pixel values increase significantly.
- Large pixel values increase only slightly.

- Hidden details in dark regions become visible.

Advantages

- Enhances poorly illuminated images.
- Improves visibility of shadow regions.
- Compresses large intensity ranges.
- Used in medical imaging and remote sensing.

Conclusion

Log transformation mainly improves low-intensity regions while preventing bright regions from becoming overexposed.

(b) Compute the transformed values.

Since the numerical values (pixel values and constant c) are missing

Example

Assume

- c = 1
- r = 15

$$\begin{aligned} s &= \log(1 + 15) \\ &= \log(16) \\ &= 2.77 \end{aligned}$$

Similarly,

For

r = 31

$$s = \log(32) = 3.46$$

**Question 3 – Histogram Processing**

(a) Explain the role of histogram equalization in contrast enhancement.

Definition

Histogram Equalization redistributes the intensity values of an image so that all grey levels are used more uniformly.

Purpose

- Improves image contrast.
- Reveals hidden details.

- Spreads frequently occurring intensity values.
- Makes the histogram approximately uniform.

Advantages

- Better brightness distribution.
- Enhanced visibility.
- Improves medical and satellite images.
- Useful before segmentation.

Applications

- X-ray enhancement
- CCTV footage
- Fingerprint recognition
- Face recognition

Conclusion

Histogram equalization automatically improves image contrast without changing image contents.

(b) Compute cumulative distribution and equalized intensity values.

The uploaded document provides only a partial histogram, and the complete frequency table is missing.

Method

Suppose

| Intensity | Frequency |
|---|---|
| 0 | 2 |
| 1 | 2 |
| 2 | 4 |
| 3 | 8 |

Step 1 – Total Pixels

$$N = 2 + 2 + 4 + 8 = 16$$

Step 2 – Cumulative Distribution

| Intensity | Frequency | CDF |
|---|---|---|
| 0 | 2 | 2 |
| 1 | 2 | 4 |
| 2 | 4 | 8 |
| 3 | 8 | 16 |

Step 3 – Equalization Formula

$$S_k = \frac{CDF}{N} \times (L - 1)$$

For a 3-bit image

$$L = 8$$

Calculations

Intensity 0

$$\frac{2}{16} \times 7 = 0.875 \approx 1$$

Intensity 1

$$\frac{4}{16} \times 7 = 1.75 \approx 2$$

Intensity 2

$$\frac{8}{16} \times 7 = 3.5 \approx 4$$

Intensity 3

$$\frac{16}{16} \times 7 = 7$$

Equalized Values

| Original | Equalized |
|---|---|
| 0 | 1 |
| 1 | 2 |

| | |
|---|---|
| 2 | 4 |
| 3 | 7 |

**Question 4 – Spatial Filtering (Smoothing)**

(a) Explain the purpose of spatial filtering for smoothing.

Definition

Spatial smoothing is performed by replacing each pixel with the average of its neighboring pixels.

Purpose

- Removes random noise.
- Reduces unwanted intensity variations.
- Produces smoother images.
- Helps segmentation algorithms.
- Removes salt-and-pepper noise.

Applications

- Medical imaging
- CCTV images
- Face recognition
- Satellite images

Advantages

- Simple implementation
- Noise reduction
- Better segmentation performance

Conclusion

Smoothing improves image quality by reducing noise while preserving the overall structure.

(b) Apply a 3×3 averaging filter.

The 3×3 pixel matrix is not visible in the uploaded document, so the exact center value cannot be computed.

Method

Average Filter

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Suppose

$$\begin{bmatrix} 20 & 30 & 40 \\ 50 & 60 & 70 \\ 80 & 90 & 100 \end{bmatrix}$$

Sum

$$20 + 30 + 40 + 50 + 60 + 70 + 80 + 90 + 100 = 540$$

Average

$$540/9 = 60$$

Final Center Pixel

60

**Question 5 – Spatial Filtering (Sharpening)**

(a) Explain how sharpening enhances image details.

Definition

Sharpening increases the visibility of edges and fine details by emphasizing rapid intensity changes.

Purpose

- Highlights edges.
- Improves object boundaries.
- Increases image clarity.
- Enhances fine textures.

Common Methods

- Laplacian filter
- High-pass filter
- Unsharp masking

Applications

- Medical diagnosis
- Object detection
- Face recognition
- Satellite image analysis

Conclusion

Sharpening makes image details more prominent by enhancing high-frequency components.

(b) Apply the Laplacian mask. Compute the output at the center.

The Laplacian mask and image matrix are missing in the uploaded document, so the exact numerical result cannot be computed.

Standard Laplacian Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Formula

$$Output = (4 \times Center) - (Top + Bottom + Left + Right)$$

Example

Suppose

$$\begin{bmatrix} 20 & 30 & 40 \\ 50 & 60 & 70 \\ 80 & 90 & 100 \end{bmatrix}$$

Center = 60

$$\begin{aligned} &= (4 \times 60) - (30 + 50 + 70 + 90) \\ &= 240 - 240 = 0 \end{aligned}$$

Interpretation

A larger positive or negative output indicates a stronger edge, while values near zero indicate a smooth region.

**Question 6 – Frequency Domain Filtering**

(a) Differentiate Low-Pass and High-Pass Filtering in the Frequency Domain

Definition

Frequency domain filtering is an image enhancement technique performed after converting an image into the frequency domain using the Fourier Transform. Different filters are applied to either preserve or remove certain frequency components.

Low-Pass Filter (LPF)

A Low-Pass Filter allows low-frequency components to pass while blocking high-frequency components.

Characteristics

- Removes noise.
- Produces a smoother image.

- Blurs sharp edges.
- Reduces fine details.

Applications

- Noise removal
- Medical image enhancement
- Preprocessing before segmentation

High-Pass Filter (HPF)

A High-Pass Filter allows high-frequency components to pass while removing low-frequency components.

Characteristics

- Sharpens the image.
- Enhances edges.
- Highlights fine details.
- Increases contrast around object boundaries.

Applications

- Edge detection
- Object recognition
- Image sharpening

Comparison Table

| Low-Pass Filter | High-Pass Filter |
|---|---|
| Passes low frequencies | Passes high frequencies |
| Removes noise | Enhances edges |
| Produces smooth image | Produces sharp image |
| Reduces details | Highlights details |
| Used for smoothing | Used for sharpening |

Conclusion

Low-pass filtering is mainly used for smoothing and noise removal, whereas high-pass filtering is used for sharpening and enhancing image details.

(b) Apply a low-pass filter that retains only the first two frequency components. Write the filtered output.

Method

Suppose the frequency components are

$$[40, \; ,25, \; ,15 \;\; 8 \;\; 5]$$

After applying a Low-Pass Filter,

Only the first two components remain.

Filtered Output

$$[40, \; ,25, \; ,0 \;\; 0 \;\; 0]$$

Interpretation

The image becomes smoother because high-frequency details such as edges and noise are removed.

**Question 7 – Thresholding**

(a) Explain Global Thresholding in Segmentation

Definition

Thresholding is one of the simplest image segmentation techniques. It separates an image into foreground and background using a threshold value.

Formula

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

where

- **T** = Threshold
- **1** = Object
- 0 = Background

Advantages

- Simple implementation
- Fast computation
- Easy object extraction
- Suitable for images with uniform lighting

Applications

- OCR (Optical Character Recognition)

- Medical imaging
- Industrial inspection
- Document scanning

Conclusion

Global thresholding converts grayscale images into binary images, making object detection easier.

(b) Apply threshold T. Assign values (0 or 1).

.

Example

Assume

Threshold = 100

Pixel values

60, 90, 120, 180

Applying threshold

| Pixel | Output |
|---|---|
| 60 | 0 |
| 90 | 0 |
| 120 | 1 |
| 180 | 1 |

Final Output

[0, 0, 1, 1]

Interpretation

Pixels above the threshold are classified as foreground (1), while pixels below the threshold are classified as background (0).

**Question 8 – Edge Detection**

(a) Explain the Concept of Edge Detection

Definition

Edge detection identifies locations where pixel intensity changes sharply. These changes represent object boundaries.

Purpose

- Detect object boundaries.

- Identify shapes.
- Improve segmentation.
- Assist in object recognition.

Common Edge Detection Operators

- Sobel
- Prewitt
- Roberts
- Canny
- Laplacian

Applications

- Face recognition
- Medical imaging
- Autonomous vehicles
- Satellite image analysis

Advantages

- Highlights boundaries.
- Reduces unnecessary information.
- Improves image analysis.

Conclusion

Edge detection extracts important structural information by identifying significant intensity changes.

(b) Apply the Horizontal Sobel Operator

The uploaded assessment references the Horizontal Sobel Operator, but the kernel and input matrix are not visible, so the exact gradient cannot be calculated.

Horizontal Sobel Mask

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Example

Input

$$\begin{bmatrix} 20 & 20 & 20 \\ 50 & 50 & 50 \\ 80 & 80 & 80 \end{bmatrix}$$

Gradient

$$\begin{aligned} &= (-20 - 40 - 20) + (80 + 160 + 80) \\ &= -80 + 320 \\ &= 240 \end{aligned}$$

Final Answer

Gradient = 240

Interpretation

A high gradient value indicates the presence of a strong horizontal edge.

**Question 9 – Region-Based Segmentation**

(a) Explain Region Growing in Segmentation

Definition

Region Growing is a segmentation technique that begins with a selected seed pixel and expands the region by including neighboring pixels with similar intensity values.

Working Steps

1. Select a seed pixel.
2. Compare neighboring pixels.
3. If the intensity difference is within the threshold, include the pixel.
4. Repeat until no more pixels satisfy the condition.

Advantages

- Produces connected regions.
- Easy implementation.
- Suitable for medical images.
- Preserves object shape.

Applications

- Tumor detection
- Satellite images
- Industrial inspection

Conclusion

Region growing groups similar neighboring pixels into meaningful image regions.

(b) Seed = 50, Threshold Difference = 5

The assessment states Seed = 50 and Threshold Difference = 5, but the pixel intensity values are missing from the uploaded document, so the exact region cannot be identified.

Method

Accept pixels satisfying

$$| Pixel - 50 | \leq 5$$

Suppose pixel values are

45, 48, 50, 52, 55, 61

Accepted pixels

48, 50, 52, 55

Rejected

45, 61

Final Region

48, 50, 52, 55

**Question 10 – Combined Enhancement & Segmentation**

(a) Explain Why Smoothing is Applied Before Segmentation

Definition

Smoothing removes random noise before segmentation.

Reasons

- Removes unwanted variations.
- Prevents false edge detection.
- Produces more accurate segmentation.
- Improves thresholding.
- Enhances object boundaries after noise removal.

Advantages

- Better object extraction.
- Reduced segmentation errors.
- Improved image quality.
- More reliable analysis.

Applications

- Medical imaging
- Face recognition
- Machine vision
- Satellite image analysis

Conclusion

Smoothing improves segmentation accuracy by reducing noise while preserving important image structures.

(b) Apply Averaging (Mean = 48), Then Threshold

The uploaded document specifies mean = 48, but the threshold value and the original image values are not visible, so the exact segmented output cannot be computed.

Example

Suppose the smoothed pixel values are

40, 48, 52, 60

Threshold = 50

| Pixel | Output |
|---|---|
| 40 | 0 |
| 48 | 0 |
| 52 | 1 |
| 60 | 1 |

Final Segmented Output

[0, 0, 1, 1]

Interpretation

Pixels greater than or equal to the threshold are segmented as the foreground (1), while the remaining pixels become the background (0).

**Question 11 – Power-Law (Gamma) Transformation**

(a) Interpret the role of gamma correction in image enhancement.

Definition

Power-law (Gamma) Transformation is a point processing technique that adjusts the brightness of an image by applying a nonlinear relationship between the input and output pixel values.

Formula

$$s = cr^{\gamma}$$

Where:

- **r** = Input pixel value
- **s** = Output pixel value
- **c** = Constant
- **γ (gamma)** = Gamma value

Role of Gamma Correction

- Controls image brightness.
- Enhances dark or bright regions.
- Improves visual quality.
- Corrects display device characteristics.
- Preserves image details.

Applications

- Medical imaging
- Digital cameras
- Computer graphics
- Satellite image enhancement

Conclusion

Gamma correction improves image visibility by controlling brightness without significantly affecting image details.

(b) Justify the use of Power-Law Transformation

When $\gamma < 1$

- Image becomes brighter.
- Dark regions become more visible.
- Low-intensity pixels are enhanced.

When $\gamma > 1$

- Image becomes darker.
- Bright regions are compressed.
- Overexposed images are improved.

Advantages

- Flexible brightness adjustment.
- Better visual perception.
- Natural image enhancement.
- Widely used in display systems.

**Conclusion**

Power-law transformation provides better brightness control than simple linear transformations.

(c) Apply the transformation step-by-step

The assessment asks for step-by-step calculations, but the required pixel values and constants are not visible in the uploaded document.

Example

Assume

- c = 1
- γ = 0.5
- Pixel value = 64

$$s = 64^{0.5}$$
$$= 8$$

Similarly,

For

r =100

$$s = \sqrt{100} = 10$$

(d) Compute the transformed values accurately

Example Table

| Input Pixel | Output Pixel (γ = 0.5) |
|---|---|
| 25 | 5 |
| 64 | 8 |
| 100 | 10 |
| 144 | 12 |

(e) Interpret the results in terms of brightness enhancement

- Dark regions become brighter.
- Hidden details become visible.
- Image contrast improves.
- Overall appearance becomes clearer.
- Useful for poorly illuminated images.

Final Conclusion

Gamma correction enhances image brightness while preserving image quality.

**Question 12 – Contrast Stretching**

(a) Interpret the need for Contrast Stretching

Definition

Contrast stretching increases the range of intensity values in an image.

Need

- Improves poor contrast.
- Enhances object visibility.
- Makes dark areas brighter.
- Makes bright areas more distinguishable.

Applications

- Medical images
- Satellite images

  Industrial inspection
- Computer Vision

(b) Justify Linear Transformation to range [0,255]

Formula

$$S = \frac{(r - r_{min})}{(r_{max} - r_{min})} \times 255$$

Where

- $r_{min} = 50$
- $r_{max} = 150$

Advantages

- Utilizes the full intensity range.
- Improves visibility.
- Enhances contrast.
- Preserves image information.

(c) Apply transformation step-by-step

Example

Input = 75

$$S = \frac{75 - 50}{150 - 50} \times 255$$
$$= \frac{25}{100} \times 255$$
$$= 63.75$$
$$\approx 64$$

(d) Compute output values

| Input | Output |
|---|---|
| 50 | 0 |
| 75 | 64 |
| 100 | 128 |
| 125 | 191 |
| 150 | 255 |

(e) Interpret improvement in contrast

- Full grayscale range is utilized.
- Dark pixels become darker.
- Bright pixels become brighter.
- Image details become clearer.
- Visual quality improves.

**Question 13 – Histogram Equalization (4-bit Image)**

(a) Interpret the need for Equalization

Histogram Equalization redistributes intensity values to improve image contrast.

Benefits

- Better brightness distribution.
- Improves visibility.
- Reveals hidden details.
- Enhances image quality.

(b) Justify use of CDF

CDF (Cumulative Distribution Function)

- Maps old intensities to new ones.
- Ensures gradual brightness change.
- Produces better contrast.
- Uses all available gray levels.

(c) Compute Cumulative Distribution

Given

| Intensity | Frequency |
|---|---|
| 0 | 4 |
| 1 | 6 |
| 2 | 10 |
| 3 | 6 |

Total Pixels

$$N = 4 + 6 + 10 + 6 = 26$$

| Intensity | Frequency | CDF |
|---|---|---|
| 0 | 4 | 4 |
| 1 | 6 | 10 |
| 2 | 10 | 20 |
| 3 | 6 | 26 |

(d) Compute New Intensity Levels

Since it is a 4-bit image

$$L = 16$$

Formula

$$S = \frac{CDF}{N} \times (L - 1)$$

Calculations

Intensity 0

$$= \frac{4}{26} \times 15 = 2.31 \approx 2$$

Intensity 1

$$= \frac{10}{26} \times 15 = 5.77 \approx 6$$

Intensity 2

$$= \frac{20}{26} \times 15 = 11.54 \approx 12$$

Intensity 3

$$= \frac{26}{26} \times 15 = 15$$

Equalized Values

| Original | Equalized |
|---|---|
| 0 | 2 |
| 1 | 6 |
| 2 | 12 |
| 3 | 15 |

(e) Interpret Contrast Enhancement

- Contrast increases.
- Histogram becomes more uniform.
- Hidden features become visible.
- Image appears clearer.

**Question 14 – Median Filtering**

(a) Interpret Noise Characteristics

Salt-and-pepper noise appears as random black and white pixels caused by transmission or sensor errors.

(b) Justify Median Filter Usage

Median filtering

- Removes impulse noise.
- Preserves edges.
- Does not blur the image.
- Better than averaging for salt-and-pepper noise.

(c) Apply Median Filter

Since the matrix is missing, consider the example below.

Window

$$[10, \ ,12, \ ,255, \ ,11, \ ,13 \ 14 \ 12 \ 10 \ 11]$$

Sorted

$$[10,10,11,11,12,12,13,14,255]$$

(d) Compute Center Value

Median

$$= 12$$

Final Answer

Center pixel = 12

(e) Interpret Noise Removal

- Impulse noise is removed.
- Edge details are preserved.
- Image quality improves.
- Suitable for medical and satellite images.

**Question 15 – Gaussian Smoothing**

(a) Interpret Smoothing Objective

Gaussian smoothing reduces random noise while preserving important image structures.

(b) Justify Gaussian Filter over Averaging

Advantages

- Gives higher weight to nearby pixels.
- Produces natural smoothing.
- Preserves edges better.
- Reduces Gaussian noise effectively.

(c) Apply Gaussian Filter

A standard Gaussian kernel is

$$\frac{1}{16}\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Since the matrix is missing in the source, the exact calculation cannot be performed.

(d) Compute Center Pixel

Example

Suppose

$$\begin{bmatrix} 20 & 20 & 20 \\ 20 & 40 & 20 \\ 20 & 20 & 20 \end{bmatrix}$$

Weighted Sum

$$= (20 + 40 + 20 + 40 + 160 + 40 + 20 + 40 + 20) = 400$$

Divide by 16

$$400/16 = 25$$

Final Center Pixel

25

(e) Interpret Smoothing Effect

- Noise is reduced.
- Image becomes smoother.
- Fine details are preserved better than with an averaging filter.
- Segmentation accuracy improves.
- The image is visually clearer.

**Question 16 – High-Pass Filtering**

(a) Interpret the Need for High-Pass Filtering

Definition

High-pass filtering is an image enhancement technique that emphasizes high-frequency components such as edges, fine details, and object boundaries while suppressing low-frequency components.

Need

- Enhances image sharpness.
- Highlights object boundaries.
- Improves visibility of fine details.
- Removes blurred appearance.
- Improves feature extraction for Computer Vision.

Applications

- Medical image analysis
- Face recognition
- Satellite image processing
- Industrial inspection

Conclusion

High-pass filtering is mainly used to sharpen images by emphasizing rapid intensity changes.

(b) Justify Filter Selection

High-pass filters are selected because they:

- Detect edges effectively.
- Increase image sharpness.
- Preserve important image details.

- Improve object recognition.
- Enhance high-frequency information.

Advantages

- Better edge visibility.
- Improved feature extraction.
- Suitable for image sharpening.
- Fast implementation.

(c) Apply the High-Pass Mask

Example High-Pass Mask

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Suppose the image matrix is

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 50 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

(d) Compute the Center Value

Using the above mask:

$$\begin{aligned} &= (8 \times 50) - (10 + 20 + 10 + 20 + 20 + 10 + 20 + 10) \\ &= 400 - 120 \\ &= 280 \end{aligned}$$

Final Answer

Center pixel = 280

(e) Interpret Edge Enhancement

- Higher output values indicate stronger edges.
- Image becomes sharper.
- Object boundaries become clearer.
- Fine details become more visible.

**Question 17 – Frequency Domain High-Pass Filtering**

(a) Interpret High-Frequency Importance

High-frequency components represent:

- Edges
- Fine textures
- Object boundaries
- Rapid intensity variations

These are important for image analysis and object detection.

(b) Justify High-Pass Filtering

High-pass filtering:

- Removes smooth background information.
- Preserves edges.
- Enhances image details.
- Improves segmentation accuracy.

(c) Apply High-Pass Filtering (Remove Low Frequencies)

Since the actual frequency values are missing, consider the example below.

Original Frequencies

$$[80,\ ,60,\ ,40\ 20\ 10]$$

After removing low frequencies

$$[0,\ ,0,\ ,40\ 20\ 10]$$

(d) Compute Output

Filtered Output

$$[0,\ ,0,\ ,40\ 20\ 10]$$

(e) Interpret Result

- Background becomes less prominent.

- Edges become sharper.
- Fine details are enhanced.
- Image appears clearer.

**Question 18 – Adaptive Thresholding**

The assessment provides the pixel values [60, 80, 120, 140] and specifies a local threshold (mean = 100).

(a) Interpret Segmentation Challenge

In images with uneven lighting, a single global threshold may not correctly separate foreground and background.

Adaptive thresholding overcomes this limitation by calculating different thresholds for different regions of the image.

(b) Justify Adaptive Thresholding

Advantages:

- Works well under non-uniform illumination.
- Produces accurate segmentation.
- Handles varying brightness.
- Reduces segmentation errors.

(c) Apply Local Threshold (Mean = 100)

Rule

If Pixel ≥ 100 → 1

Otherwise → 0

(d) Compute Binary Output

Given pixels

$$[60, \ ,80, \ 120 \ 140]$$

| Pixel | Binary Output |
|---|---|
| 60 | 0 |

| | |
|---|---|
| 80 | 0 |
| 120 | 1 |
| 140 | 1 |

Final Output

$$[0, \ ,0, \ 1 \ 1]$$

(e) Interpret Result

- Pixels below the threshold become background.
- Pixels above the threshold become foreground.
- Object separation becomes easier.
- Segmentation accuracy improves.

**Question 19 – Edge Detection (Vertical Sobel)**

(a) Interpret Edge Detection Objective

Edge detection identifies significant intensity changes that represent object boundaries.

Objectives

- Detect boundaries.
- Recognize shapes.
- Improve segmentation.
- Support object recognition.

(b) Justify Operator Choice

Vertical Sobel Operator detects:

- Vertical edges.
- Sudden horizontal intensity changes.
- Object boundaries.

Advantages:

- Simple implementation.
- Noise resistant.
- Produces accurate gradients.

(c) Apply Vertical Sobel Operator

Standard Vertical Sobel Kernel

$$\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

Example Matrix

$$\begin{bmatrix} 10 & 20 & 30 \\ 10 & 20 & 30 \\ 10 & 20 & 30 \end{bmatrix}$$

(d) Compute Center Value

Calculation

$$\begin{aligned} &= (-10 + 30) + (-20 + 60) + (-10 + 30) \\ &= 20 + 40 + 20 \\ &= 80 \end{aligned}$$

Final Answer

Gradient = 80

(e) Interpret Edge Direction

- Large positive value → Strong vertical edge.
- Small value → Smooth region.
- Larger magnitude indicates stronger boundaries.

**Question 20 – Prewitt Edge Detection**

(a) Interpret Prewitt Operator Purpose

The Prewitt operator is an edge detection technique used to estimate the image gradient and identify object boundaries.

Purpose

- Detect edges.
- Highlight object boundaries.
- Improve segmentation.

- Extract structural information.

(b) Justify Use over Sobel

Compared to Sobel:

- Simpler computation.
- Faster implementation.
- Suitable for basic edge detection.
- Useful for educational and simple applications.

(c) Apply Prewitt Kernel

Example Horizontal Kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Example Matrix

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

(d) Compute Output

Calculation

$$\begin{aligned} &= (-10 - 10 - 10) + (30 + 30 + 30) \\ &= -30 + 90 \\ &= 60 \end{aligned}$$

Final Output

Gradient = 60

(e) Interpret Result

- High gradient values indicate strong edges.
- Object boundaries become clearly visible.
- Edge information is extracted for further image analysis.
- The image becomes suitable for segmentation and feature extraction.

**Question 21 – Region Splitting**

(a) Interpret the Segmentation Problem

Definition

Region splitting is a segmentation technique that divides an image into smaller regions whenever a region does not satisfy a predefined homogeneity criterion.

Need

- Separate dissimilar pixel groups.
- Improve object identification.
- Handle images with varying intensity values.
- Increase segmentation accuracy.

Applications

- Medical imaging
- Satellite image analysis
- Object detection
- Industrial inspection

Conclusion

Region splitting divides an image into meaningful regions based on intensity differences.

(b) Justify Splitting Method

The splitting method is preferred because:

- It separates pixels having large intensity differences.
- Produces homogeneous regions.
- Improves segmentation quality.
- Easy to implement.
- Suitable for images with non-uniform intensities.

(c) Apply Threshold = 80

Given Pixels

$$[40, \ ,45, \ 100 \ 105]$$

Rule

- Pixels $< 80 \rightarrow$ Region 1
- Pixels $\geq 80 \rightarrow$ Region 2

(d) Compute Regions

Region 1

$$[40, 45]$$

Region 2

$$[100, 105]$$

(e) Interpret Separation

- Two homogeneous regions are obtained.
- Low-intensity pixels belong to one object.
- High-intensity pixels belong to another object.
- Segmentation becomes more accurate.

**Question 22 – Region Merging**

(a) Interpret Merging Criteria

Definition

Region merging combines adjacent regions that have similar intensity values into a single larger region.

Purpose

- Reduce over-segmentation.
- Produce larger meaningful objects.
- Simplify image analysis.

(b) Justify Merging Approach

Region merging is useful because:

- Adjacent similar regions belong to the same object.
- Reduces unnecessary partitions.
- Improves segmentation accuracy.
- Produces connected regions.

(c) Apply Threshold = 5

Region 1

$$[45, 50]$$

Average

$$= \frac{45 + 50}{2} = 47.5$$

Region 2

$$[52, 55]$$

Average

$$= \frac{52 + 55}{2} = 53.5$$

Difference

$$53.5 - 47.5 = 6$$

The difference (6) is greater than the threshold (5).

(d) Compute Merged Region

Since the difference is greater than 5, the two regions should not be merged.

Final Regions

Region 1 = [45, 50]

Region 2 = [52, 55]

(e) Interpret Result

- The regions remain separate.
- Each region represents different intensity groups.
- Better segmentation accuracy is maintained.

**Question 23 – Edge Linking**

The assessment provides the detected edges [1, 0, 1, 0, 1].

(a) Interpret Problem of Discontinuity

Definition

Edge detection often produces broken or disconnected edges due to noise or weak gradients.

Problem

- Object boundaries become incomplete.
- Shape recognition becomes difficult.
- Segmentation accuracy decreases.

(b) Justify Linking Method

Edge linking is used because:

- Connects broken edges.
- Produces continuous boundaries.
- Improves object detection.
- Reduces fragmentation.

(c) Apply Linking Step

Given

$$[1, \ , 0, \ , 1 \ 0 \ 1]$$

Replace isolated gaps (0) between edge pixels with 1.

(d) Compute Result

Final linked edges

$$[1, \ , 1, \ , 1 \ 1 \ 1]$$

(e) Interpret Continuity

- Continuous boundaries are formed.
- Object outlines become complete.
- Feature extraction becomes easier.
- Image analysis improves.

**Question 24 – Low-Pass Filtering (Frequency Domain)**

(a) Interpret Smoothing Requirement

Low-pass filtering removes high-frequency noise and smooths the image.

Benefits

- Reduces random noise.
- Produces smoother images.
- Removes unnecessary details.
- Improves image quality.

(b) Justify LPF Selection

Advantages

- Removes high-frequency noise.
- Improves segmentation.
- Preserves overall image structure.
- Suitable for preprocessing.

(c) Apply Filter

Example

Original Frequency Components

$$[90,\ ,60,\ ,40\ \ 20\ \ 10]$$

Retain only low frequencies.

Filtered Output

$$[90,\ ,60,\ ,0\ \ 0\ \ 0]$$

(d) Compute Output

Final Frequency Components

$$[90,\ ,60,\ ,0\ \ 0\ \ 0]$$

(e) Interpret Effect

- Noise is reduced.
- Image becomes smoother.
- Sharp edges become less prominent.
- Overall image quality improves.

**Question 25 – Laplacian Sharpening (Variant)**

(a) Interpret Sharpening Need

Definition

Laplacian sharpening enhances image details by emphasizing areas where pixel intensity changes rapidly.

Need

- Improve image clarity.
- Highlight edges.
- Increase object visibility.
- Enhance fine details.

(b) Justify Mask

The Laplacian mask is selected because it:

- Detects intensity changes.
- Sharpens blurred images.
- Enhances object boundaries.
- Improves feature extraction.

(c) Apply the Mask

Standard Laplacian Mask

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Example Image

$$\begin{bmatrix} 20 & 30 & 20 \\ 30 & 50 & 30 \\ 20 & 30 & 20 \end{bmatrix}$$

(d) Compute Center Pixel

Calculation

$$= (4 \times 50) - (30 + 30 + 30 + 30)$$
$$= 200 - 120$$
$$= 80$$

Final Answer

Center Pixel = 80

(e) Interpret Result

- Edge information is enhanced.
- Image becomes sharper.
- Fine details become clearer.
- Object boundaries become more visible.

**Question 26 – Histogram Matching**

(a) Interpret the Need for Histogram Matching

Definition

Histogram Matching (Histogram Specification) is an image enhancement technique that transforms the histogram of one image so that it closely matches the histogram of a reference image.

Purpose

- Standardizes image appearance.
- Improves image comparison.
- Enhances visual consistency.
- Matches brightness and contrast between images.

Applications

- Medical imaging
- Satellite image analysis
- Face recognition
- Image databases

Conclusion

Histogram matching makes one image resemble another in terms of brightness and contrast.

(b) Justify the Method

Histogram matching is preferred because:

- It provides controlled enhancement.
- Matches images captured under different lighting conditions.
- Produces consistent intensity distribution.
- Improves image interpretation.

(c) Map Values [10, 50, 100]

The uploaded document specifies the values [10, 50, 100] but does not provide the target histogram, so exact mapping is not possible.

Example

Assume the target histogram maps:

| Original Value | Mapped Value |
|---|---|
| 10 | 20 |
| 50 | 70 |
| 100 | 140 |

(d) Compute New Values

Using the example mapping:

- $10 \rightarrow 20$
- $50 \rightarrow 70$
- $100 \rightarrow 140$

Final Output

$$[20, 70, 140]$$

(e) Interpret the Result

- Brightness becomes similar to the reference image.
- Contrast is improved.
- Images become visually consistent.
- Better comparison between images is possible.

**Question 27 – Noise Reduction Comparison**

(a) Interpret the Noise Type

Noise is an unwanted disturbance that degrades image quality.

Common Types

- Salt-and-pepper noise
- Gaussian noise
- Speckle noise
- Poisson noise

(b) Compare Averaging and Median Filters

| Averaging Filter | Median Filter |
|---|---|
| Computes the mean of neighboring pixels. | Computes the median of neighboring pixels. |
| Smooths the image. | Removes impulse noise effectively. |
| May blur edges. | Preserves edges better. |
| Suitable for Gaussian noise. | Suitable for salt-and-pepper noise. |

Conclusion

Median filtering is generally preferred for salt-and-pepper noise because it preserves edges while removing noise.

(c) Apply Both Filters

Example Window

$$[10\text{،}\ \text{،}12\text{،}\ \text{،}255\text{،}\ \text{،}11\text{،}\ \text{،}13\ \ 14\ \ 12\ \ 10\ \ 11]$$

Averaging Filter

Sum

$$10 + 12 + 255 + 11 + 13 + 14 + 12 + 10 + 11 = 348$$

Average

$$348/9 = 38.67$$

Median Filter

Sorted Values

$$10,\ 10,\ 11,\ 11,\ 12,\ 12,\ 13,\ 14,\ 255$$

Median

$$= 12$$

(d) Compute Outputs

Average Filter Output

$$39$$

Median Filter Output

$$12$$

(e) Interpret Differences

- Averaging reduces noise but blurs edges.
- Median removes impulse noise effectively.
- Median preserves important image details.
- Median filtering provides better image quality for salt-and-pepper noise.

**Question 28 – Combined Filtering**

(a) Interpret the Need for Smoothing + Sharpening

Definition

Combined filtering first removes noise using smoothing and then enhances important details using sharpening.

Need

- Remove unwanted noise.
- Improve image clarity.
- Preserve useful details.
- Enhance object boundaries.

(b) Justify the Sequence

The correct order is:

Step 1: Smoothing

- Removes random noise.

Step 2: Sharpening

- Enhances edges after noise removal.

If sharpening is applied first, the noise will also be sharpened.

(c) Apply Filters Stepwise

Example

Original Pixel

50

After Smoothing

48

After Sharpening

55

(d) Compute Output

Example Output

| Stage | Pixel Value |
|---|---|
| Original | 50 |
| After Smoothing | 48 |
| After Sharpening | 55 |

(e) Interpret Improvement

- Noise is reduced.
- Edges become sharper.
- Object boundaries are clearer.
- Image quality is significantly improved.

**Question 29 – Multi-Threshold Segmentation**

The assessment provides pixel values [20, 60, 120, 180] and thresholds 50 and 100.

(a) Interpret Multi-Level Segmentation

Definition

Multi-threshold segmentation divides an image into more than two classes using multiple threshold values.

Purpose

- Separate objects with different intensity levels.
- Improve segmentation accuracy.
- Identify multiple regions.

(b) Justify Thresholds (50,100)

The thresholds divide the image into three intensity regions:

- Low intensity
- Medium intensity
- High intensity

This provides better segmentation than using a single threshold.

(c) Apply Segmentation

Rules

- Class 1: Pixel $< 50$
- Class 2: $50 \leq$ Pixel $< 100$
- Class 3: Pixel $\geq 100$

(d) Compute Classes

| Pixel | Class |
|---|---|
| 20 | Class 1 |
| 60 | Class 2 |
| 120 | Class 3 |
| 180 | Class 3 |

Final Output

$$[C,l,a,ss\ 1\ Class\ 2\ Class\ 3\ Class\ 3]$$

(e) Interpret

- Image is divided into three meaningful regions.
- Different objects become easier to identify.
- Segmentation accuracy improves.

**Question 30 – Boundary Detection**

(a) Interpret Boundary Extraction

Definition

Boundary detection identifies the outer edges of objects in an image by detecting intensity changes.

Purpose

- Locate object outlines.
- Measure object shape.
- Assist in object recognition.
- Improve segmentation.

(b) Justify the Method

Boundary detection combines:

- Edge detection to identify object borders.
- Edge linking to connect broken edges.

This produces complete object boundaries.

(c) Apply Edge Detection + Linking

Example

Detected Edges

$$[1,\ ,0,\ ,1\ 0\ 1]$$

After Edge Linking

$$[1,\ ,1,\ ,1\ 1\ 1]$$

(d) Compute Result

Final Boundary

$$[1, \ , 1, \ , 1 \ 1 \ 1]$$

(e) Interpret Boundaries

- Continuous object boundaries are obtained.
- Object shapes become clearly visible.
- Feature extraction becomes easier.
- Image segmentation accuracy improves.